Process Characterization Report

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCR-004

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

BLA, Biologics License Application · CPP, critical process parameter · CQA, critical quality attribute · DNA, deoxyribonucleic acid · DS, drug substance · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · IPC, in-process control ·

KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · NTU, nephelometric turbidity unit · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · rcf, relative centrifugal force · SDM, scale-down model · SEC, size-exclusion chromatography.

Executive summary

This report documents the Stage 1 (Process Design) characterization of the A-Mab harvest and clarification operation (Step 4), the primary-recovery unit operation that separates the antibody-containing supernatant from the cells and debris of the production culture and delivers a clarified feed to the Protein A capture step. The study was executed under the Process Characterization Plan **PCP-004** on a qualified bench-scale clarification model, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012), with the process model and quality-attribute framework of the A-Mab case study (CMC Biotech Working Group 2009).

3 process parameters were evaluated one factor at a time. The characterization confirms that harvest and clarification has **no impact on the product-quality CQAs**, which pass through the operation unchanged, and that the operation robustly delivers a clarified feed of consistent quality to Protein A.

Key outcomes

- No product-quality CQA is set or modified by harvest and clarification; the glycan, charge-variant and aggregate attributes are established upstream and pass through unchanged, and the HCP/DNA load is formed upstream and cleared downstream.
- The clarification step yield is **97.7%** at the nominal condition (65,596 g product mass in, 64,071 g out), preserving cumulative process yield at **97.7%** through Step 4.
- The post-clarification turbidity is **5 NTU**, within its 1–10 NTU normal operating range, confirming a consistent, low-turbidity feed to Protein A.
- Centrifugation force and depth-filter load are classified **KPP** (they govern recovery and feed clarity without a CQA impact); post-clarification turbidity is a monitored **GPP**. No parameter is a CPP.

The harvest and clarification operation is demonstrated to be well understood and robust. Because it forms no product-quality attribute, its parameter ranges and control are justified on process-performance grounds, and this report rolls up into the Process Characterization Master Report (**PCMR-001**).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose product-quality attributes — Fc N-glycosylation, charge variants and aggregate — are established in the production bioreactor and controlled through the purification train (CMC Biotech Working Group 2009). Harvest and clarification is the primary-recovery operation that separates the antibody-containing supernatant from the cells and cell debris of the fed-batch culture and delivers a clarified feed to the Protein A capture step. In the A-Mab platform it comprises continuous disk-stack centrifugation followed by depth-filter and sterile-filter clarification. The operation removes cells and particulates but does not form, remove or modify any product-quality attribute; it is therefore characterized as a process-performance operation.

1.2 Objectives

The objectives of the study were to (i) confirm that harvest and clarification has no impact on the product-quality CQAs; (ii) establish the effect of the operating parameters on product yield, post-clarification turbidity and depth-filter throughput; (iii) confirm that a clarified feed of consistent quality is delivered to Protein A across the operating ranges; (iv) establish proven acceptable ranges (PARs) and normal operating ranges (NORs) for each parameter; and (v) classify each parameter and provide the ranges that justify the Stage 2 operating conditions and the control strategy.

1.3 Regulatory and scientific basis

The work follows the lifecycle approach to process validation of the FDA 2011 guidance (U.S. Food and Drug Administration 2011), PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), and applies the enhanced (QbD) development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012). Criticality is treated as a continuum (ICH Q9). The report structure corresponds to the Stage 1 Process Design Report of PDA TR 60 §3.11 and provides development-report content suitable to support a Biologics License Application; it is correspondingly focused, because the operation forms no CQA and therefore requires no designed multivariate experiment.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The characterization ranges and set-points are supported by platform knowledge from related humanized IgG1 antibodies and by A-Mab development and engineering experience (CMC Biotech Working Group 2009). Prior knowledge established that the glycan, charge-variant and aggregate CQAs are set in cell culture and that the harvest HCP and DNA burden is generated upstream and cleared by the purification train; harvest neither sets nor clears these attributes. This knowledge framed the process-performance measures in scope and the decision to characterize the operation univariately.

2.2 Quality attributes and process-performance measures

Harvest and clarification **sets or modifies no product-quality CQA**; the CQAs established in the production bioreactor pass through the operation unchanged. The operation was therefore characterized against the process-performance measures it governs (Table 2.1): product (step) yield, post-clarification turbidity (feed clarity to Protein A) and depth-filter throughput. The impurity load presented to Protein A — the harvest HCP and residual DNA formed upstream — was monitored to confirm feed consistency but is not an attribute this step controls.

Table 2.1: Process-performance measures characterized for harvest and clarification (no product-quality CQA is set by this step).

Process-performance measure	Result at nominal condition	Basis / criterion
Clarification step yield	97.7%	Mass balance across primary recovery
Post-clarification turbidity	5 NTU	Within NOR 1–10 NTU
Depth-filter throughput	Within capacity	Filter sizing / no breakthrough
Harvest HCP / residual-DNA load	Monitored (consistent)	Formed upstream; cleared downstream

2.3 Risk-based prioritization and parameter selection

Consistent with the A-Mab lifecycle of iterative risk assessments, the characterization scope was set by the Pre-Characterization Process Risk Assessment (**RA-001**). Because harvest and clarification carries no credible risk of impact to a product-quality CQA, none of its parameters was assigned to a multivariate design of experiments; each parameter was studied one factor at a time across its characterization range for its effect on the process-performance measures. The parameters and ranges carried into this study are given in Table 4.1.

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were performed on a qualified bench-scale clarification model representing the commercial primary-recovery train — a laboratory disk-stack (or equivalent bench) centrifuge operated at the commercial-equivalent relative centrifugal force and feed-solids loading, followed by scaled-down depth and sterile filters of the platform media at the commercial area-normalized load (**SOP-2005**, **SOP-2006**). The model was qualified against at-scale reference data (**SOP-1001**) by comparing centrate and filtrate turbidity, product yield and the HCP/DNA load presented to Protein A, confirming that the model is representative and suitable to support the parameter ranges (Parenteral Drug Association 2013).

3.2 Operation

Harvest was initiated on the nominal duration criterion of the production culture. The culture was clarified by continuous centrifugation to remove cells and large debris, and the centrate was passed through a depth filter and a sterile (bioburden-reduction) filter to remove residual particulates before capture. The relative centrifugal force, the area-normalized depth-filter load and the resulting post-clarification turbidity were the operating variables. Buffers, filters and media were controlled to platform specifications and treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

3.3 Analytical methods

Feed clarity and the feed impurity load were measured by validated methods (Table 3.1). Post-clarification turbidity was determined by nephelometry (**AMV-3015**); HCP by ELISA (**AMV-3012**); residual host-cell DNA by qPCR (**AMV-3014**); and size variants (aggregate/HMW) by SEC-HPLC (**AMV-3011**) to confirm the aggregate attribute is unchanged. Method performance characteristics are summarized in Appendix B.

Table 3.1: Validated analytical methods used in the characterization.

Reference	Title	Type
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

3.4 Sampling plan

Each characterization run was sampled at the clarified-pool stage for post-clarification turbidity and for the harvest HCP and residual-DNA load, and the product mass across the operation was recorded for the step-yield calculation. The depth-filter pressure/throughput profile was recorded to confirm filter capacity. Runs at the parameter set-points were replicated to quantify the reproducibility of yield and turbidity.

3.5 Statistical methods

Because no product-quality CQA is affected, the operation was characterized univariately rather than by a designed experiment. Each parameter was varied one factor at a time across its characterization range, holding the others at their set-points, and the process-performance measures — step yield, post-clarification turbidity and depth-filter throughput — were summarized against the parameter level. Reproducibility of yield and turbidity was estimated from the replicated set-point runs. Parameter ranges were judged acceptable where the performance measures remained within their platform targets and the product-quality CQAs were unchanged.

4 Study design

4.1 Parameters, ranges and the knowledge space

3 parameters were studied (Table 4.1). The characterization range of each parameter ("PAR" column) defines the edges of the explored knowledge space and the basis of the proven acceptable range; the NOR is the tighter range at which the parameter is controlled routinely. Classification is an output of the study (§8).

Table 4.1: Harvest / clarification process parameters, set-points, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Centrifugation (rcf)	g	9e+03	8000–10000	6000–12000	KPP	univariate
Depth filter load	L/m2	120	100–140	80–160	KPP	univariate
Post-clarification turbidity	NTU	5	1–10	1–20	GPP	univariate

4.2 Univariate ranging

Each parameter was evaluated one factor at a time across its characterization range; none was carried into a multivariate design because prior knowledge and the Pre-Characterization Risk Assessment indicated no credible CQA impact. The product- quality CQAs were confirmed unchanged across the ranging.

5 Results

5.1 Clarification performance and reproducibility

At the nominal (set-point) condition the operation recovered antibody with a clarification step yield of 97.7% (65,596 g product mass in, 64,071 g out), preserving the cumulative process yield at 97.7% through Step 4. The post-clarification turbidity was 5 NTU, within its 1–10 NTU normal operating range, and the depth filter operated within its rated capacity without particulate breakthrough. The set-point replicates confirmed that yield and turbidity are reproducible, establishing that the operation delivers a consistent clarified feed to Protein A.

5.2 Univariate ranging of the operating parameters

Across the characterization range of each parameter, the process-performance measures remained within their platform targets and the product-quality CQAs were unchanged:

- **Centrifugation (relative centrifugal force).** Increasing the relative centrifugal force across its range improves centrate clarity and reduces the solids load on the depth filter, at the cost of a higher shear/lysis burden; product yield and feed clarity remained acceptable across the range, and the NOR was confirmed as the routine-control window that balances clarity against cell lysis.
- **Depth-filter load (area-normalized).** Increasing the area-normalized load reduces the required filter area but raises filtrate turbidity and the risk of particulate breakthrough and pressure-driven yield loss near capacity; turbidity and yield remained within target across the range, and the NOR was confirmed as the window that preserves filter capacity and feed clarity.
- **Post-clarification turbidity.** As the controlled output attribute of the operation, post-clarification turbidity remained within its normal operating range across the operating conditions, confirming a consistent, low-turbidity feed to Protein A.

5.3 No product-quality impact

The size-variant (aggregate) result confirmed that the aggregate attribute is unchanged across the operation, and the glycan and charge-variant attributes — set in cell culture — are not modified by clarification. The harvest HCP and residual-DNA load, formed upstream, was consistent run-to-run and is cleared by the purification train. The characterization therefore confirms that harvest and clarification has no product-quality impact, consistent with the A-Mab case study (CMC Biotech Working Group 2009).

6 Process performance and consistency

Because the operation forms no product-quality CQA, its robustness is expressed as the consistency of its process-performance measures rather than as a multivariate design space or a CQA capability index. The clarification yield of 97.7% and the post-clarification turbidity of 5 NTU, both reproducible at the set-point and stable across the parameter ranges, establish that the operation reliably delivers a clarified feed of consistent quality to Protein A at commercial scale. The commercial-scale CQA capability is reported for the CQA-forming and CQA-clearing unit operations in their respective reports and consolidated in **PCMR-001**; harvest contributes to it only by preserving yield and feed consistency.

7 Parameter classification

Each parameter was classified on the A-Mab continuum from the demonstrated effects on process performance and the absence of a CQA impact (Table 7.1). The rationale is as follows.

- **Centrifugation force, depth-filter load — KPP.** Each governs recovery, centrate/ filtrate clarity and filter capacity (process performance and feed consistency) but does not significantly impact any CQA; they are controlled for process performance.
- **Post-clarification turbidity — GPP.** A monitored output attribute of the operation with a wide acceptable range and no product-quality consequence; it is trended to confirm feed consistency.

No harvest parameter is a CPP, because the operation forms no product-quality attribute.

Table 7.1: Post-characterization parameter classification.

Parameter	Class	Study
Centrifugation (rcf)	KPP	univariate
Depth filter load	KPP	univariate
Post-clarification turbidity	GPP	univariate

8 Contribution to the control strategy

Harvest and clarification is not a control point for any product-quality CQA; its contribution to the overall control strategy is to protect process performance and feed consistency. That contribution comprises: control of the centrifugation force and depth-filter load to their NORs for reproducible recovery and clarity; in-process monitoring of post-clarification turbidity as the feed-clarity attribute presented to Protein A; trending of the harvest HCP and residual-DNA load to confirm the impurity load presented to the purification train is consistent; and the sterile (bioburden-reduction) filtration that protects the downstream capture step. The CQAs that load the purification train — HCP, DNA and aggregate — are set upstream and cleared by the Protein A, cation-exchange and anion-exchange steps (reported in the paired unit-operation reports and consolidated in **PCMR-001**).

9 Discussion

The characterization confirms the expected role of harvest and clarification: a primary-recovery operation that defines the feed to Protein A without forming or modifying any product-quality attribute. The operating parameters influence recovery and feed clarity in the directions established by platform knowledge, and their ranges are justified on process-performance grounds. The principal limitation is inherent to scale-down characterization: absolute yield, centrate clarity and the harvest impurity load may shift modestly at commercial scale, which is why the model is confirmed at scale during Stage 2 and the KPP ranges are carried forward with in-process monitoring of turbidity and the feed impurity load. The characterization did not identify any parameter requiring critical (CPP) control within the studied ranges.

10 Conclusions

The A-Mab harvest and clarification operation is well understood and robust. It sets or modifies no product-quality CQA; it recovers antibody at a clarification yield of 97.7% and delivers a clarified feed at 5 NTU within its normal operating range; and its parameters are classified as KPP or GPP with a data-supported rationale. The parameter ranges and classifications provide the basis for the Stage 2 operating ranges and the process-performance elements of the drug-substance control strategy. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

Deviations from the plan

No deviations from **PCP-004** were recorded. All planned ranging runs were executed and all planned measures were recorded on the qualified scale-down clarification model.

References

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Appendix A — Univariate ranging summary

The parameters, characterization ranges and the process-performance measures recorded at each level. Each parameter was varied one factor at a time with the remaining parameters held at their set-points.

Table 10.1: Appendix A — one-factor-at-a-time ranging summary.

Parameter	Set-point	PAR	NOR	Class	Performance measures recorded
Centrifugation (rcf)	9e+03	6000–12000	8000–10000	KPP	step yield, post-clarification turbidity, depth-filter throughput
Depth filter load	120	80–160	100–140	KPP	step yield, post-clarification turbidity, depth-filter throughput
Post-clarification turbidity	5	1–20	1–10	GPP	step yield, post-clarification turbidity, depth-filter throughput

Appendix B — Analytical methods summary

Table 10.2: Analytical methods, techniques and validation references (performance characteristics per the cited method validation reports).

Method	Technique	Analytes / measures	Validation ref.	Reportable
Turbidity	Nephelometry	Post-clarification turbidity	AMV-3015	NTU
Host-cell protein	ELISA	HCP (feed load)	AMV-3012	ng/mg
Residual DNA	qPCR	Host-cell DNA (feed load)	AMV-3014	ng/dose
Size variants	SEC-HPLC	Aggregate (HMW), monomer	AMV-3011	% HMW